

Designing a Multi-Hole Pitot Probe for UAV Measurements in Dust-Devil Flows

Introduction

Dust devils are compact vortices that loft dust and particulates from the surface on Earth and Mars. They influence dust transport and local radiative forcing, yet we still lack high-resolution, *in situ* measurements of their internal velocity and pressure structure. Tower-based sensors cannot intercept the core, and satellites only retrieve column-integrated signatures. My long-term goal is to close this observational gap by developing flight-ready instrumentation for fixed-wing UAVs capable of safely transecting dust devils.

Over the past year, I designed, built, and began calibrating a five-hole Pitot probe and electronics pod for reconstructing the 3-D flow field in real time. This project became my first full experience taking an instrument from conceptual aerodynamics to embedded systems to wind-tunnel calibration, and it taught me how to reason across physical and engineering constraints.

Methods

Probe Design and Fabrication. I began by studying multi-hole probe theory and the UK National Physical Laboratory (NPL) “Modified Ellipsoidal” Pitot design [1], which uses a 4:1 ellipsoidal nose to minimize suction peaks and maintain attached flow. Following these guidelines, I designed a 5 mm OD stainless-steel stem with a resin-printed ellipsoidal head truncated to a flat tip ratio of $d/D \approx 0.16$, which is within the recommended 0.15–0.25 range that keeps the suction peak below $\sim 10\%$ of dynamic pressure [2].

I embedded five 0.5 mm OD aluminum microtubes into the head, developing a repeatable cutting and alignment method to avoid ovalizing or blocking the ports. This procedure is now the lab’s build reference for new

probes. Downstream, I designed transition cones and a sealed reference manifold that routes the three lower ports to a common chamber. Using ideal-gas and stagnation–static relations, I confirmed that although the internal reference pressure drifts with altitude, differential pressures such as $P_0 - P$ remain accurate because all ports share a common offset [3].



Figure 1: Five-hole Pitot probe with ellipsoidal nose, flat tip, and embedded microports.

From Pressure to Velocity. To verify sensor suitability, I first estimated dynamic pressures using the incompressible relation $\Delta P = \frac{1}{2}\rho V^2$. For 10–25 m/s, dynamic pressures span 60–240 Pa, well matched to the ± 250 Pa range of the differential sensors. Because dust-devil transects may approach $M \approx 0.2$, I checked that compressibility effects would remain small by evaluating the isentropic stagnation–static relation [4]

$$\frac{P_0}{P} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\gamma/(\gamma-1)},$$

which confirmed that deviations from incompressible behavior are $< 1\%$ across our operating regime.

In the calibration pipeline, port pressures are normalized using the stagnation–static span ($P_{\max} - P_{\min}$), which represents the dynamic-pressure range resolved by the five ports. Forming left–right and top–bottom differences and dividing by this span produces the standard non-dimensional coefficients C_α , C_β , and C_q used in multi-hole probe calibration. Working through this procedure clarified how these normalized pressure differences encode angle of attack, sideslip, and velocity magnitude, and how the pitch–yaw sweep data generate smooth calibration surfaces.

Tubing Losses and Geometry. Tubing bends introduce pressure losses with coefficients ranging from $K \approx 0.2$ for long-radius elbows to $K \approx 1.0$ for sharp 90° turns, as summarized in EPA and NPL tables [5]. To ensure that calibration remains valid during flight, I fix the tubing geometry on the wing and calibrate with the same routing so that all geometric losses are absorbed directly into the calibration map.

Wind-Tunnel Calibration Setup. I mounted the probe on a pitch–yaw rig and acquired preliminary data in a closed-section wind tunnel. I swept yaw and pitch angles while recording all five differential pressures across multiple freestream speeds. These measurements form the basis of the multidimensional calibration surface, following the NPL-L formalism.

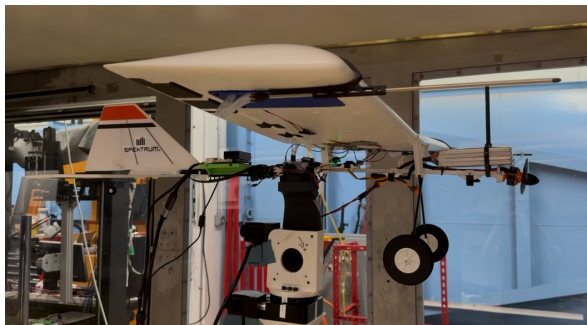


Figure 2: Wind-tunnel configuration for multi-hole probe calibration using a pitch–yaw rig and fixed tubing geometry.

Embedded Systems Integration. In parallel, I assembled the embedded sensing pipeline. I configured the Teensy firmware to read the five differential-pressure sensors over I²C through a multiplexer and checked that the data streams were correctly formatted and time-stamped. On the Jetson Nano, I set up ROS Noetic and MAVROS, established the serial links to the Cube Orange+ autopilot, and built the launch structure that ingests the Teensy data and synchronizes them with the UAV state estimate.

I also configured ROS time as the system clock so that pressure, IMU, and autopilot messages align consistently. For telemetry, I routed MAVLink through MAVProxy to QGroundControl on a NUC ground station, since Qt6/ARM64 incompatibilities prevent QGroundControl from running natively on the Jetson.

Results

By the end of the integration phase, the full sensing system was ready for controlled free-flight testing. I completed all wiring and soldering for the pressure-sensor board, multiplexer, Teensy, and Jetson interfaces, and verified electrical continuity and stable sensor reads under repeated power cycles. I also enabled a transmitter-based manual control mode so that the pilot can immediately take over if the sensing or estimation pipeline behaves unexpectedly.

To ensure that the software infrastructure behaved reliably, I ran the complete ROS–MAVLink stack with the autopilot, monitoring message timing, synchronization, and link stability. These tests confirmed that the pressure data, IMU orientation, and UAV state remain synchronized within a shared ROS time base, that MAVLink routes cleanly through MAVProxy, and that the system recovers gracefully if any single component is restarted.

Taken together, these integration results demonstrate that the sensing pipeline functions as a cohesive, flight-ready system capable of supporting free-flight experiments that combine multi-hole pressure measurements with full-state UAV navigation.

Conclusion

This project taught me how to integrate aerodynamics, instrumentation, and embedded computing into a flight-ready system. I learned how to interpret historical aerodynamic designs, fabricate precise probe geometries, debug dense sen-

sor networks, model compressible Pitot behavior, and construct robust ROS/MAVLink pipelines. Most importantly, I learned to reason like an experimentalist by identifying confounding factors, quantify uncertainties, and validate every assumption with calibration data.

Future Studies

Dust devils exhibit strong electric-field amplification driven by triboelectric charging and vertical charge separation [6, 7]. I plan to integrate a compact E-field probe alongside the Pitot system to measure polarity and magnitude of these gradients. Combining these measurements with flow vectors will allow tests of hypotheses about whether stronger vortices generate faster charge growth through increased grain collisions.

Dust devils also exhibit asymmetric temperature gradients across the vortex front and rear [8]. By co-locating a microbolometer or IR sensor with the Pitot probe, I can measure these gradients directly and relate them to dynamic pressure deficits. Martian analog studies show that near-surface thermal inversions may drive vortex formation [9], and IR measurements may help discriminate environmental regimes.

Finally, I plan to integrate a lightweight RGB or multispectral camera and use structure-from-motion (SfM) photogrammetry to reconstruct dust distributions. UAVs overcome many limitations of satellites such as fixed viewing geometry, low resolution, meteorological dependence by providing multi-angle, close-proximity imaging [10]. SfM-MVS techniques [11] will enable 3-D reconstruction of dust concentration and vortex-wall thickness.

Taken together, these extensions will allow this instrument to evolve into a multi-modal platform capable of probing the coupled thermodynamic, electrostatic, and

aerodynamic structure of dust devils. I hope to deepen this foundation throughout my future research career.

References

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